

AZTIBASE NETWORK

The AI-Native Blockchain

Business in a Box

Product strategy for plug-and-play validator hardware — turning a blockchain node into an appliance anyone can operate.

\$699

Standard Node

\$1,500

AI Compute Node

\$200+

Margin / Unit

7 min

Unbox to Earning

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01 Executive Summary

Aztibase Network is a Layer-1 AI-native blockchain where validators earn rewards by running useful AI inference instead of solving meaningless hash puzzles. The protocol is production-ready: 44,900 lines of Rust across 9 crates, 940+ passing tests, and a live 3-node testnet producing 400ms blocks with sub-second finality.

The **Business in a Box** model transforms this technology into a physical product: a pre-configured mini PC that arrives ready to validate. The operator plugs it in, connects to WiFi, enters an invite code in a browser-based setup wizard, and begins earning AZTB tokens within minutes. No terminal, no compilation, no DevOps knowledge required.

This document defines two hardware SKUs, their unit economics, the customer journey from unboxing to earning, the technical architecture that enables plug-and-play operation, and the phased fulfillment strategy from white-label to custom OEM.

Core thesis: *The validator hardware business generates immediate fiat revenue (\$200+ margin per unit), seeds the network with real validators, and creates a self-reinforcing flywheel — more validators increase network security, which increases token value, which increases hardware demand.*

02 Product Definition — Two SKUs

Two distinct products serve two distinct buyer profiles. The Standard Node targets the crypto-curious operator who wants passive income from staking. The AI Compute Node targets the technically ambitious operator who wants to earn additional PoUW inference rewards.

SKU Comparison

	Standard Node	AI Compute Node
Target Buyer	Passive staker, non-technical	Power user, AI-curious operator
Base Hardware	Beelink SER5 (Ryzen 5, 16GB)	Beelink SER7 (Ryzen 7, 32GB)
CPU	6-core / 12-thread	8-core / 16-thread
RAM	16 GB DDR4	32 GB DDR5
Storage	500 GB NVMe	1 TB NVMe
Network	1 GbE + WiFi 6	2.5 GbE + WiFi 6E

	Standard Node	AI Compute Node
AI Capability	None (PassthroughRuntime)	64 MiB ONNX inference via tract
Earnings	Staking rewards (70% of emission)	Staking + PoUW rewards (70% + 15%)
Retail Price	\$699	\$1,499
Hardware Cost	~\$280	~\$520
Gross Margin	~\$419 (60%)	~\$979 (65%)

What's in the Box

Item	Standard Node	AI Compute Node
Pre-configured mini PC	✓	✓
Pre-flashed Ubuntu + Aztibase image	✓	✓
Power adapter + ethernet cable	✓	✓
Quick-start card with invite code	✓	✓
Aztibase logo sticker / branded sleeve	✓	✓
Genesis validator slot	✓	✓
3 reference ONNX models pre-loaded	—	✓
Extended setup support (Discord priority)	—	✓

03 Unit Economics & Bill of Materials

Standard Node — BOM at 50 Units

Component	Unit Cost	Notes
Beelink SER5 5560U (16GB / 500GB)	\$230	Bulk pricing via AliExpress / direct
NVMe pre-flash (Ubuntu + Aztibase)	\$5	SD card cloner + labor
Logo sticker + branded sleeve	\$3	Print run of 100+
Quick-start card (printed)	\$1	Double-sided card stock
Ethernet cable (1m Cat6)	\$2	Bulk pack
Packaging (kraft box + foam insert)	\$8	Custom cut foam
QA testing + burn-in (30 min/unit)	\$15	Labor cost
Shipping to customer (domestic avg)	\$15	Courier, insured
Total COGS	\$279	
Retail Price	\$699	
Gross Margin	\$420 (60%)	

AI Compute Node — BOM at 50 Units

Component	Unit Cost	Notes
Beelink SER7 7840HS (32GB / 1TB)	\$500	Direct or Amazon bulk
NVMe pre-flash + ONNX model bundle	\$5	Includes 3 reference models
Logo sticker + branded sleeve	\$3	Same print run
Quick-start card (printed)	\$1	
Ethernet cable (1m Cat6)	\$2	
Packaging (kraft box + foam insert)	\$10	Slightly larger box
QA testing + burn-in (30 min/unit)	\$15	
Shipping to customer (domestic avg)	\$15	
Total COGS	\$551	
Retail Price	\$1,499	

Component	Unit Cost	Notes
Gross Margin	\$948 (63%)	

Margin at Scale

Volume	Std Node COGS	Std Margin	AI Node COGS	AI Margin
50 units	\$279	\$420 (60%)	\$551	\$948 (63%)
100 units	\$255	\$444 (64%)	\$510	\$989 (66%)
500 units	\$220	\$479 (69%)	\$450	\$1,049 (70%)
1,000 units (OEM)	\$180	\$519 (74%)	\$380	\$1,119 (75%)

Break-even analysis: At 50 units of Standard Node (\$699 each), total revenue is \$34,950 with \$20,950 gross profit. Fixed costs (OS image development, web UI, packaging design) are estimated at ~\$2,000. The hardware business is profitable from the first batch.

04 Customer Journey — Unbox to Earning

The entire experience is designed around a single constraint: the operator should never open a terminal. Every interaction happens through the physical hardware and a browser.

1. UNBOX

Open box. Remove mini PC, power adapter, ethernet cable, and quick-start card.

2. CONNECT

Plug in power + ethernet (or WiFi via WPS button). The node boots automatically.

3. DISCOVER

Open any browser on the same network. Navigate to **http://aztibase.local** (mDNS). The setup wizard greets you.

4. NAME

"Name your node" — enter a friendly name (e.g., "My Validator"). This sets the node identity on the network.

5. SECURE

Set a 6-digit PIN. Behind the scenes: Ed25519 + BLS validator keys are generated and encrypted with the PIN. Optional: write down 24-word recovery phrase.

6. JOIN

Enter the network invite code from the quick-start card (e.g., **AZTB-7K3M-QXPW**). The code decodes to genesis hash + boot node addresses. The node downloads the genesis config and connects to peers automatically.

7. SYNC

Progress screen: "Finding peers... Syncing blocks... Ready!" — typically under 60 seconds on a fresh network.

8. EARN

Dashboard appears: node status, block height, peer count, balance, and earnings. The operator is now a validator, earning AZTB every epoch (~7 minutes).

Total time from opening the box to seeing the first reward: under 7 minutes. No Rust toolchain. No git clone. No CLI flags. No SSH. No cloud accounts. This is the DappNode experience but lighter, cheaper, and AI-native.

05 Technical Architecture — The 4-Layer Stack

The plug-and-play experience requires four software layers, each eliminating a category of technical friction. All layers ship with the hardware — no external services required.

Layer	What It Eliminates	Technology	Effort
1. Pre-Flashed OS Image	Compile, install, configure Linux	Ubuntu Server minimal + systemd + avahi (mDNS)	1–2 days
2. Web Setup Wizard	CLI key generation, TOML editing, flag memorization	Rust axum server, htmx + vanilla JS, 5 screens	3–5 days
3. Built-In Dashboard	curl, jq, Prometheus, Grafana	Same axum server, real-time via SSE, RPC-backed panels	2–3 days
4. Auto-Updater	SSH maintenance, manual binary swaps	Systemd timer, Ed25519-signed binaries, rollback support	1–2 days

Layer 1 — Pre-Flashed OS Image

Each unit ships with an NVMe drive pre-loaded with Ubuntu Server 24.04 LTS minimal. The Aztibase binary is pre-compiled and installed at `/usr/local/bin/aztibase`. A systemd service (`aztibase-validator.service`) auto-starts the node on boot. Avahi daemon enables mDNS discovery so the box is reachable at `aztibase.local` from any device on the LAN. No terminal access is needed — ever.

Layer 2 — Web Setup Wizard

The existing axum HTTP server (which already serves the JSON-RPC API on port 9944) gains a set of HTML routes for the setup wizard. This means zero additional processes, zero Docker, zero nginx. The wizard is 5 screens: Welcome, Network (invite code), Security (PIN + optional mnemonic backup), Connecting (progress), and Dashboard (redirect). All key generation happens server-side in Rust using the existing wallet module.

Layer 3 — Built-In Dashboard

Panel	Data Source (RPC)	Update Frequency
Node Status (online / syncing / error)	aztb_nodeInfo	5 seconds
Block Height	aztb_blockHeight	Per block (~400ms)
Peer Count	aztb_nodeInfo → peer_count	10 seconds
Your Balance	aztb_getBalance	Per epoch (~7 min)
Earnings Over Time	Local balance delta tracking	Per epoch

Panel	Data Source (RPC)	Update Frequency
Network Validators	aztb_getActiveValidators	Per epoch
Node Uptime	Local timestamp	Continuous

Layer 4 — Auto-Updater

A systemd timer checks for new releases once daily from a configured update endpoint. The update payload is an Ed25519-signed binary. The updater verifies the signature, swaps the binary atomically (rename, not overwrite), and restarts the service. If the new binary fails health checks within 60 seconds, automatic rollback to the previous version. The operator sees an 'Update available' banner on the dashboard, or auto-applies if configured.

The Invite Code System

Instead of sharing genesis.toml files and multiaddr strings, the network coordinator generates a short, human-readable invite code:

AZTB-7K3M-QXPW

This encodes (via base32): the genesis block hash (network identity), one or more boot node multiaddrs, and an optional network display name. The new validator enters this single code in the web wizard. The box resolves everything automatically — zero copy-paste of hex strings, zero TOML editing, zero chance of joining the wrong network.

06 Fulfillment Model — Phase 1 & Phase 2

Phase 1 — White-Label (0–100 units)

Buy off-the-shelf Beelink mini PCs, flash them in-house, add branded packaging. This requires zero upfront tooling investment and can begin as soon as the web UI ships.

Step	Action	Cost	Time
1	Order 10–50 Beelink units (AliExpress / Amazon)	\$2,300–\$14,500	1–2 weeks
2	Create master OS image (Ubuntu + Aztibase + wizard)	\$0 (dev time)	2–3 days
3	Flash NVMe via USB cloner (10–15 min/unit)	\$200 (cloner hardware)	1–2 days
4	QA test: boot, wizard, join testnet, verify dashboard	\$0 (labor)	30 min/unit
5	Package: kraft box, foam, quick-start card, sticker	\$12/unit	Batch
6	Ship to customer	\$15/unit avg	2–5 days

Phase 2 — Custom OEM (100+ units)

Once 100+ white-label units are sold and the product-market fit is validated, move to custom-branded hardware from Shenzhen OEM manufacturers. This improves margins, strengthens brand identity, and enables custom enclosure design.

Aspect	Details
Manufacturers	Jinghong (odminipc.com), OAI PC, INCTEL, Shenzhen JIALAIBAO
Minimum Order	100 units (some accept 50)
Customization	Branded enclosure, logo, custom BIOS splash, pre-installed OS image, packaging
Per-Unit Cost (100)	\$200–\$350 (vs \$280 white-label)
Per-Unit Cost (500)	\$150–\$250
Lead Time	45–90 days from order to delivery
Upfront Investment	\$20K–\$35K for 100 units
Enclosure Options	CNC machined (\$20–50/unit, \$2K tooling) or injection molded (\$2–5/unit, \$15K tooling)

Phase 1 validates demand with zero tooling risk. Phase 2 only triggers after selling 100+ units proves the market. Presale revenue from Phase 1 can fund the Phase 2 OEM order.

07 Revenue Model — Stacked Income Streams

The Business in a Box model generates revenue from four complementary streams. Hardware sales provide immediate fiat revenue. Protocol fees generate recurring token income. The combination creates a self-reinforcing flywheel.

Stream	Revenue Type	Margin	When
Hardware Sales	Fiat (one-time)	60–75%	Immediate
Validator NFTs	Crypto (one-time)	~95% (digital)	Pre-mainnet
Protocol Fees	AZTB (recurring)	5% of all PoUW inference fees	Post-mainnet
VaaS Commission	AZTB (recurring)	5–15% of delegator rewards	Post-mainnet

Year 1 Revenue Projection (Conservative)

Product	Units / Events	Price	Revenue
Standard Node	50–100 units	\$699	\$35K–\$70K
AI Compute Node	10–20 units	\$1,499	\$15K–\$30K
Validator NFTs (limited 1,000)	200–500 mints	\$200–\$1,000	\$100K–\$250K
Token Sale (Fjord Foundry LBP)	1 event	—	\$100K–\$300K
Total Year 1			\$250K–\$650K

The Flywheel Effect

More hardware sold → more validators → more network security → higher token value → more attractive staking rewards → more hardware demand. Each unit sold strengthens the network, which makes the next unit easier to sell. This is the same dynamic that powered Helium's growth to 1M+ hotspots.

Validator Earnings Model

Year 1–2 emission: 120M AZTB/year. Validator share: 70% = 84M AZTB. PoUW share: 15% = 18M AZTB. At various network sizes:

Validators	Staking Reward / Validator / Year	+ PoUW (AI Nodes)	Total / Validator
10	8,400,000 AZTB	1,800,000 AZTB	10,200,000 AZTB

Validators	Staking Reward / Validator / Year	+ PoUW (AI Nodes)	Total / Validator
50	1,680,000 AZTB	360,000 AZTB	2,040,000 AZTB
100	840,000 AZTB	180,000 AZTB	1,020,000 AZTB
500	168,000 AZTB	36,000 AZTB	204,000 AZTB

Note: Actual USD value depends on AZTB market price. At \$0.01/AZTB, a validator in a 100-node network earns ~\$10,200/year. At \$0.10/AZTB, that becomes ~\$102,000/year. The hardware pays for itself quickly in most scenarios.

08 Go / No-Go Checklist

All items must be checked before ordering the first hardware batch. These are hard prerequisites — shipping hardware without them creates support debt and reputational risk.

#	Prerequisite	Status	Blocks
1	Web setup wizard (5 screens) functional on testnet	Not started	Hardware v1
2	Built-in dashboard showing status, balance, peers, earnings	Not started	Hardware v1
3	Invite code system (encode/decode/resolve)	Not started	Hardware v1
4	Pre-flash OS image tested on target hardware (Beelink SER5)	Not started	Hardware v1
5	Auto-updater with signed binary verification	Not started	Hardware v1.1
6	Friends testnet stable with 5+ validators for 7+ days	In progress	Presale launch
7	Security audit (at least consensus + wallet crates)	Planned	Mainnet
8	Legal opinion from SA crypto attorney (CASP classification)	Not started	Token sale
9	Presale page live on aztibase.com with payment processing	Not started	Presale launch
10	Support channel (Discord) with FAQ and troubleshooting guide	Partial	Hardware v1

09 Risk Matrix

Risk	Likelihood	Impact	Mitigation
Hardware defects / DOA units	Medium	High	QA burn-in test every unit. Keep 5% buffer stock. Beelink has 1-year warranty.
Low demand (< 20 units sold)	Medium	Medium	Phase 1 is white-label — no upfront tooling. Unsold units become team/testnet nodes.
Token price near zero at launch	Medium	High	Hardware margin provides fiat revenue regardless of token price. Business survives at \$0 AZTB.
Regulatory action (SA CASP)	Low	High	Engage attorney pre-launch. Hardware sales are not crypto services. Token entity offshore if needed.
Solo developer risk	High	High	Recruit 2–3 advisors. Document everything (59 sprints, full build log). Code is the proof.
Supply chain disruption	Low	Medium	Multiple OEM suppliers identified. Beelink available from Amazon, AliExpress, direct.
Competitor launches similar product	Low	Low	44,900 lines of production code is a deep moat. No other L1 has AI at consensus level.
Support burden overwhelms solo dev	Medium	Medium	Web UI eliminates 90% of support queries. FAQ + Discord community handles the rest.

10 Timeline & Milestones

Phase	Timeline	Deliverables	Revenue Event
Friends Testnet	Now – Month 1	5+ validators running, stability proven, feedback collected	—
Web UI Development	Month 1–2	Setup wizard, dashboard, invite codes, auto-updater	—
Hardware Prototype	Month 2	5 units flashed + tested on Beelink SER5, dogfooded internally	—
Presale Launch	Month 3	aztibase.com presale page, Early Bird tier (50 units @ \$599)	\$30K–\$50K
First Batch Ships	Month 4	50 Standard Nodes delivered, onboarding support	\$35K
Validator NFT Mint	Month 4–5	Limited 1,000 NFTs granting mainnet validator slots	\$100K–\$250K
Mainnet Genesis	Month 6–7	Genesis ceremony with hardware validators, token goes live	—
Token Sale (LBP)	Month 7–8	Fjord Foundry LBP, DEX liquidity seeded	\$100K–\$300K
AI Compute Node Launch	Month 8–9	Second SKU ships, PoUW inference tasks live on mainnet	\$15K–\$30K
OEM Transition	Month 10–12	Custom-branded hardware, improved margins, scale to 500+ units	Ongoing

The Business in a Box model turns Aztibase from a software project into a hardware product company with immediate fiat revenue. The validator hardware generates \$200–\$950 margin per unit, seeds the network with real participants, and creates a flywheel where every unit sold makes the network more valuable. Combined with Validator NFTs and a token launch, the conservative Year 1 projection is \$250K–\$650K — more than enough to fund engineering, legal, and infrastructure for the next phase of growth.

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Infrastructure for distributed intelligence.